

Project Snapshot :

This project delivers an end-to-end Customer Intelligence framework for an E-Commerce Subscription platform. By combining automated data processing in **KNIME Analytics Platform** with interactive modeling in **Power BI**, the project analyzes customer transaction logs to segment subscribers based on **Recency, Frequency, and Monetary (RFM)** behavior

Tools Used :

Microsoft Excel: Raw data inspection, missing value handling, structural sanitization.

KNIME Analytics Platform: Advanced ETL pipeline automation, date arithmetic, group-by aggregations, numeric binning, and rule-based RFM scoring.

Power BI Desktop: Relational modeling, DAX measures, custom visuals, dynamic page navigation, analytics quadrant styling.

Key Insights:

Value Customers dominate the user base: Representing 42.75% of customers (171 users), this group generates the highest total volume (\$6.025K aggregate revenue) with an average order spend of \$15.04.

Champions demonstrate top account value: 79 customers (19.75% of total) average \$37.78 in spend per account, exhibiting low recency and high overall engagement.

Low Monetary / High Frequency Discrepancy: A distinct cohort of subscribers orders frequently (4-5 orders) but spends under \$15 per order.

Microsoft Excel: Raw data inspection, missing value handling, structural sanitization.

KNIME Analytics Platform: Advanced ETL pipeline automation, date arithmetic, group-by aggregations, numeric binning, and rule

Choose Customer Category

About to Sleep

At Risk

Champions

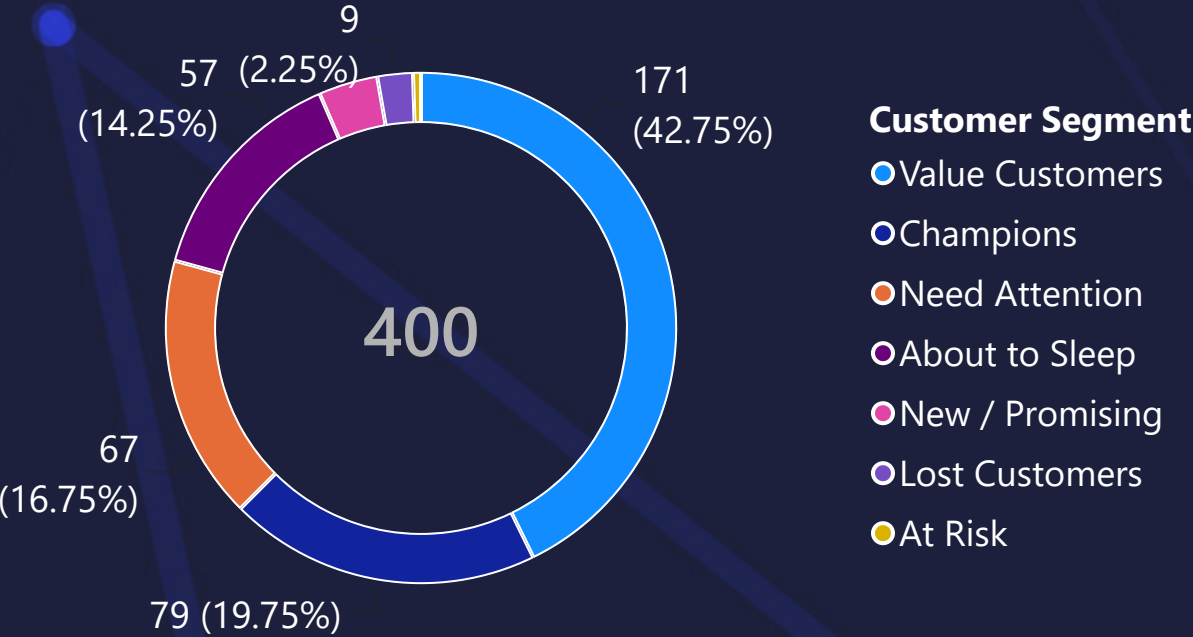
Lost Customers

Need Attention

New / Promising

Value Customers

Customer Segmentation



At Risk Customers/May Become inactive

Customer Segment	Sum of Moneytory_Value	Count of Recency_Value	customer_id
At Risk	\$17.521	1	368
At Risk	\$27.1531	1	404
Total	\$44.6741	2	

Total Revenue

6.02K

Total Customer

400

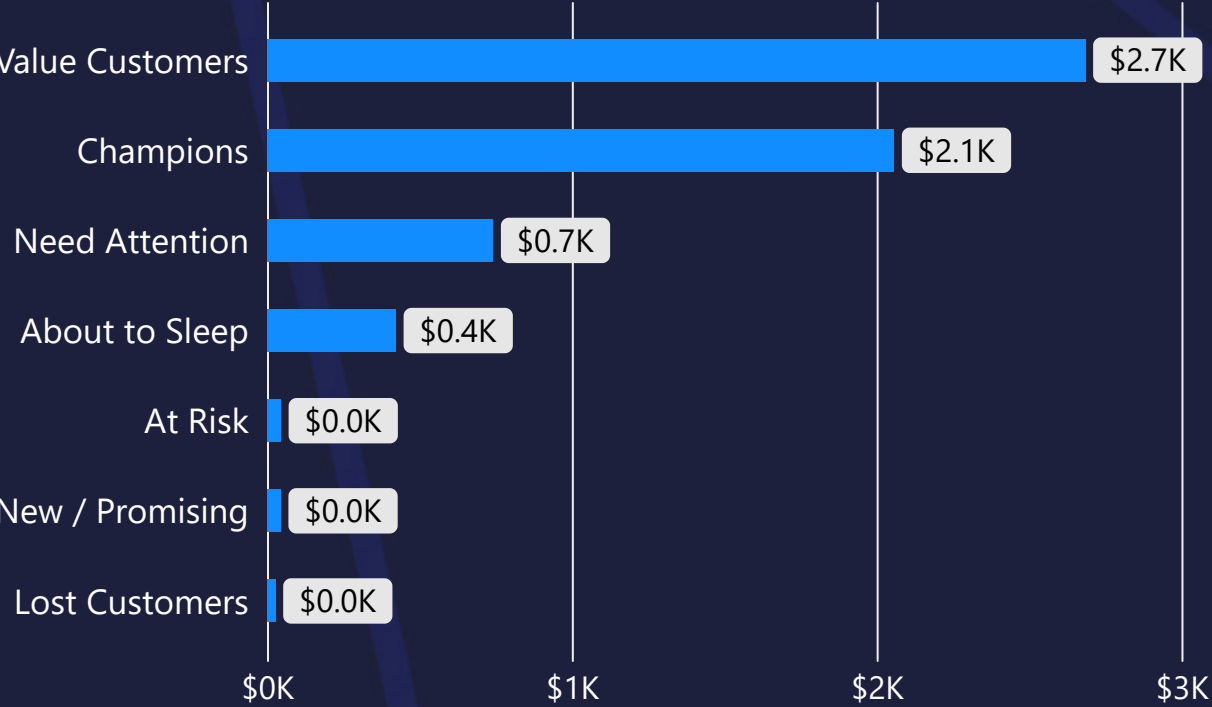
Average Sales Per Customer

\$15.0477

Total Transactions

1221

Revenue Generation by Customers Groups



Top 10 Highest Revenue Generating Customers

customer_id	Sum of Moneytory_Value
474	\$37.9459
283	\$37.7778
104	\$35.833
401	\$35.1656
222	\$34.7974
124	\$34.3256
342	\$34.1901
472	\$33.9089
265	\$33.8615
495	\$33.8595
Total	\$351.6653

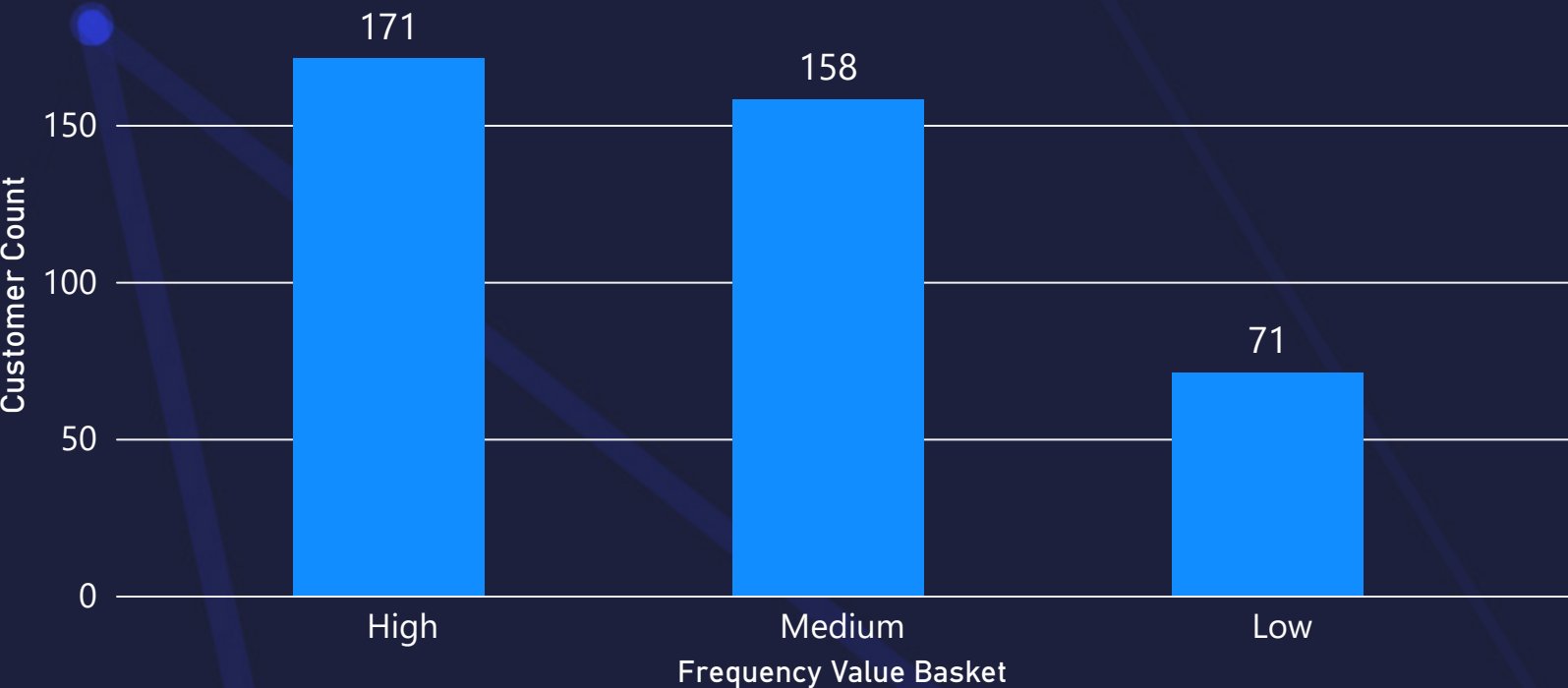
Customer Segmentations



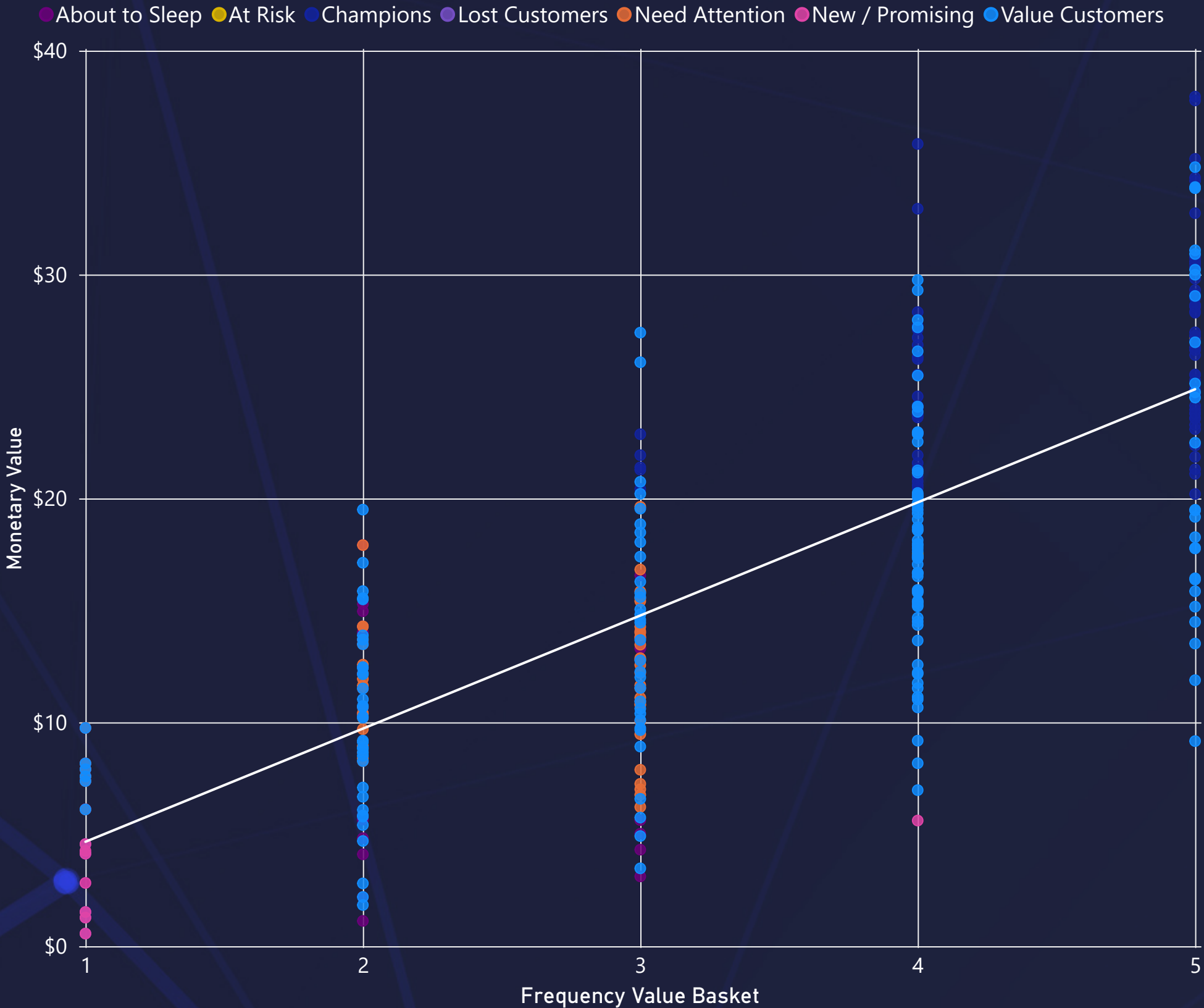
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- At Risk
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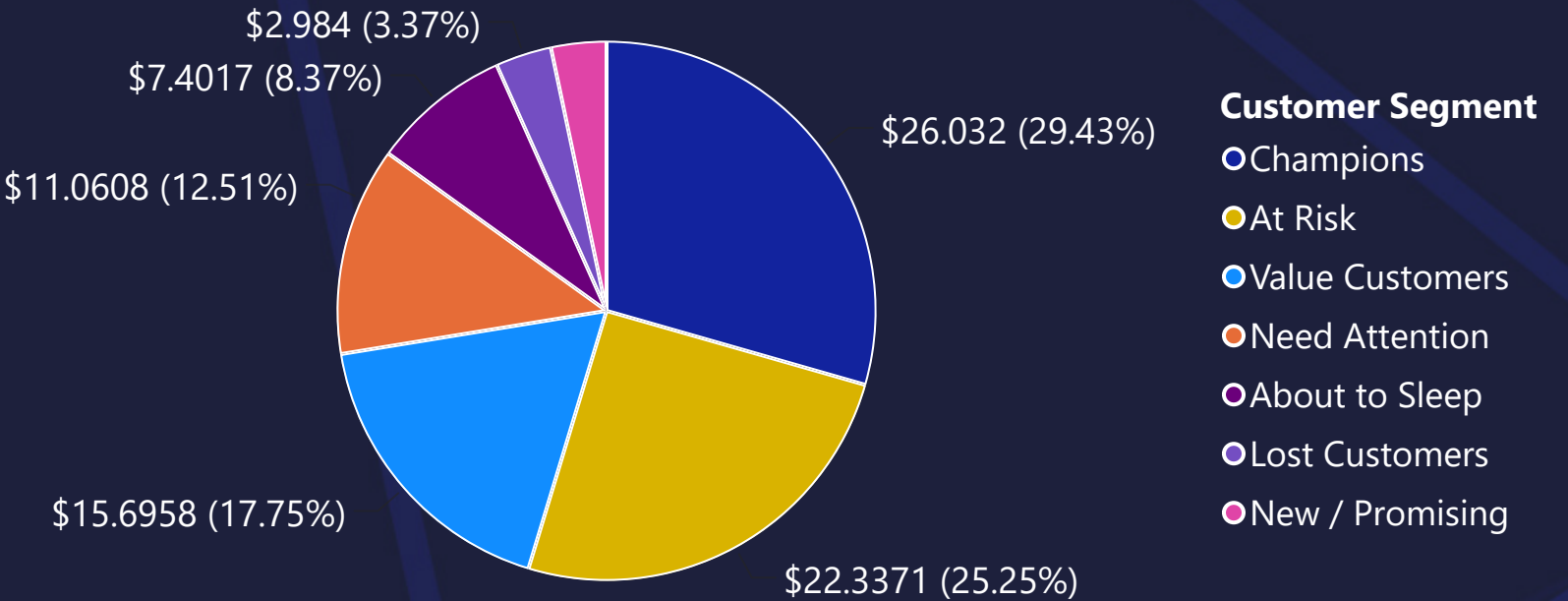
Frequency Analysis



High Frequency Low Monetary Value



Segments With Highest Average Value



Recent Buyers With High Potential

Frequency_Value	Moneytory_Value	Recency_Value	Customer Segment	customer_id
2	\$10.684	276	Value Customers	482
2	\$10.7325	297	Value Customers	119
2	\$11.029	266	Value Customers	442
2	\$12.4677	266	Value Customers	340
2	\$13.6962	268	Value Customers	303
2	\$15.4912	286	Value Customers	447
2	\$15.5441	293	Value Customers	327
2	\$17.1278	265	Value Customers	318
2	\$19.5002	266	Value Customers	248